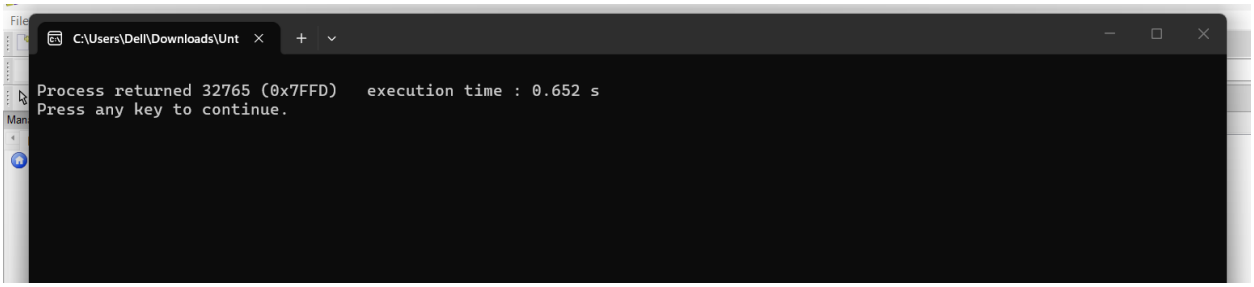
EXP:22

CODE:

```c
#include <stdio.h>

#define MAX_MEMORY 1000

int memory[MAX_MEMORY];

// Function to initialize memory
void initializeMemory() {
   for (int i = 0; i < MAX_MEMORY; i++) {
      memory[i] = -1; // -1 means free
   }
}

// Function to display memory status
void displayMemory() {
   int i, j;
   int count = 0;

   printf("\nMemory Status:\n");

   for (i = 0; i < MAX_MEMORY; i++) {
      if (memory[i] == -1) {
         count++;
         j = i;

         while (j < MAX_MEMORY && memory[j] == -1) {
            j++;
         }

         printf("Free memory block %d - %d\n", i, j - 1);
         i = j - 1;
      }
   }

   if (count == 0) {
      printf("No free memory available.\n");
   }
}
```

```c
// Function to allocate memory ( ⚠ currently FIRST FIT, not worst-fit)
void allocateMemory(int processId, int size) {
   int start = -1;
   int blockSize = 0;

   for (int i = 0; i < MAX_MEMORY; i++) {
      if (memory[i] == -1) {
         if (blockSize == 0) {
            start = i;
         }
         blockSize++;
      } else {
         blockSize = 0;
      }

      if (blockSize >= size) {
         break;
      }
   }

   if (blockSize >= size) {
      for (int i = start; i < start + size; i++) {
         memory[i] = processId;
      }

      printf("Allocated memory block %d - %d to Process %d\n",
            start, start + size - 1, processId);
   } else {
      printf("Memory allocation failed for Process %d\n", processId);
   }
}

// Function to deallocate memory
void deallocateMemory(int processId) {
   for (int i = 0; i < MAX_MEMORY; i++) {
      if (memory[i] == processId) {
         memory[i] = -1;
      }
   }

   printf("Memory released by Process %d\n", processId);
}

int main() {
   initializeMemory();
   displayMemory();

   allocateMemory(1, 200);
   displayMemory();

   allocateMemory(2, 300);
   displayMemory();

   deallocateMemory(1);
   displayMemory();

   allocateMemory(3, 400);
   displayMemory();

   return 0;
}
```

OUTPUT:


C:\Users\Dell\Downloads\Unt
Process returned 32765 (0x7FFD) execution time : 0.652 s
Press any key to continue.